

EEE 419/591

Summ 2026

Project 3

Introduction

Let's first consider the most basic diode problem. We can model a diode with this equation

$$I = I_s \left(e^{\frac{qV_d}{nkT}} - 1 \right) \quad (1)$$

where I is the current flowing in the diode in Amps, V_d is the diode voltage in Volts, T is the room temperature in Kelvins, while I_s , q , n , k and T are diode-specific fixed parameters. Fig. 1 plots the diode current versus its voltage for a different set of parameter values. These plots are for illustration purposes only. You should not rely on them to estimate the values of any of the parameters in this project. Notice that the y-axis is a log scale!

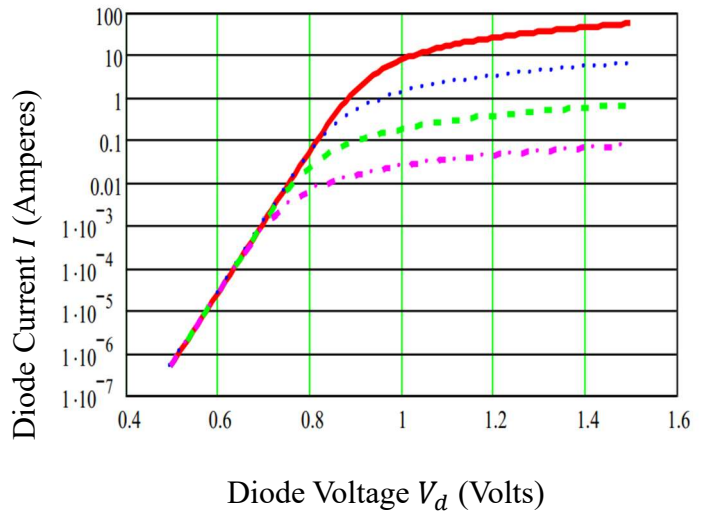


Figure 1: Diode Current versus its voltage

The two main unknowns in equation (1) are I and V_d . Knowing one yields the other.

However, the complication comes when you place this diode in a circuit where both are unknown. Consider the case where the diode is in series with a resistor R and a DC voltage source V (Fig. 2). The current voltage relationship is KVL

$$V = IR + V_d \quad (2)$$

Solving equations (1) and (2) via substitution yields a single non-linear equation in a single unknown. The main goal in this project is two-fold:

1. Learning how to use Python to solve such a non-linear equation while knowing the diode parameters.
2. Learning how to use that equation to estimate the diode parameters, using Python, given some noisy circuit measurements.

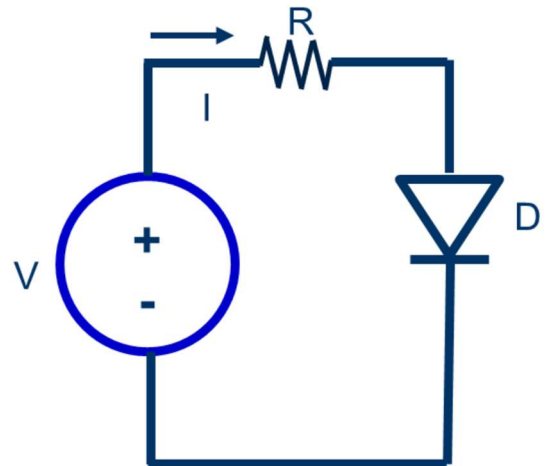


Figure 2: Circuit used for Problems 1-2

Problem 1

1. Substitute the diode current, I , in equation (1) into equation (2). On a piece of paper, write down the resulting equation. Call it equation (3). Its only variable is the diode voltage, V_d . Make sure that all the terms in equation (3) are on the right side. The left side MUST be a zero.
2. Write down a Python code that uses `optimize.fsolve` to solve for V_d to satisfy such an equation. This Python solver takes a function, iterates over its first variable and terminates when that variable results in the function being zero. Hopefully, you now understand why equation (3) must have a zero left-side.
3. Use these parameters in your simulation: $I_s = 1 \times 10^{-9} \text{ Amps}$, $n = 1.7$, $R = 11k\Omega$, $T = 350^\circ \text{ K}$ and applied voltage source V from 0.1 to 2.5 V. Analyze with steps of 0.1 V.

Problem 1 deliverables: Your code should output one plot with one curve: (a) log(Diode Current) vs Diode Voltage. This curve should be a straight-line following equation (1).

Problem 2

In this problem, we will assume that we connected the circuit in Fig. 2 with a potentially different diode than that in Problem 1. We know the values of all diode parameters but one, the ideality parameter, n . We varied the voltage source and measured the corresponding current value I . You are supplied this noisy dataset in the file named *DiodeIV.txt*. Your objective is to estimate the ideality parameter, n .

You will need to use a Python package like `optimize.leastsq`. This function is called as follows:

`optimize.leastsq(opt_n, n_init, args=(R, phi, P2_AREA, P2_T, source_v, meas_diode_i))`. It takes a function handler `opt_n`, an initial guess `n_init`, as well as the circuit parameters `args`. A part of the code is written for you in `project3.py`. You will need to finish that code and submit your `project3.py` file which should also include the code for Problem 1. The bulk of your work for this problem will be to write the function `opt_n`.

Dataset Properties:

1. It has two columns:
 - a. Column 1: voltage source values in volts
 - b. Column 2: measured diode current (noisy) values in Amperes
2. All the measurements in the dataset you are given are taken for the same diode. Hence, they were all obtained with the same ideality parameter n .
3. The following parameter values were used to generate this dataset:
 $k = 1.380648 \times 10^{-23}$ (Boltzmann constant)
 $q = 1.6021766208 \times 10^{-19}$ (electron charge in Coulombs)

$\phi = 0.89$ (barrier height)

$R = 1,150 \, \Omega$ (resistor's resistance)

$I_s = AT^2 e^{-\frac{\phi q}{kT}}$ (Saturation current)

$A = 10^{-8} \, m^2$ (Diode cross sectional area)

$T = 375^\circ K$ (room temperature)

4. You must use the data type ***float64*** when reading the dataset. Otherwise, you won't capture all decimal points. As you will see, some current measurements are in the order of 10^{-14} Amperes. The ***read_csv*** method in pandas defaults to ***float64***. If you use ***numpy***'s ***loadtxt*** method, on the contrary, make sure to set the ***dtype*** to ***float64***.

Properties of the ***opt_n*** function:

1. Inputs: all circuit parameters as inputs and the measured diode currents from the dataset. The first of these inputs **MUST** be the parameter you are optimizing over, ***n***.
2. Outputs: returns a normalized array of errors each represents the difference between the calculated current value, from equation (1), and the measured value, from the dataset.
3. Each error value corresponds to one of the measurements in the given dataset.
4. If the circuit parameters you pass to ***opt_n*** match those of the diode used to obtain the measurements in the provided dataset, ***opt_n*** returns an array of almost all-zeros. If one of the parameters does not match, the returned errors array would have at least one non-close-to-zero value.
5. Calculating the current from equation (1) requires solving the circuit using equation (3). Hence, ***opt_n*** can call functions you wrote for Problem 1 above.
6. Normalization of errors: normalization of the errors array is strongly recommended or otherwise an error of $1e-13$ Amperes in a current of $1e-14$ Amperes would be neglected compared to an error of $1e-4$ in a current of $1e-3$ Amperes even though the former is an error of 10x while the latter is of 0.1x only. The normalization equation is arbitrary and is up to you to figure. You are allowed to share your normalization equation with your classmates over Ed Discussion and use the best one you find.

Properties of ***optimiz.leastsq***:

1. It works as follows: It starts with the initial guess substitutes in the ***opt_n*** function, receives the corresponding returned error array, and if it is not zero, it fine-tunes the initial guess and repeats this process until it terminates when the function ***opt_n*** returns a zero (or a vector of zeros). Since the data is noisy, an exact zero is not achievable.
2. It can be passed a function that returns a vector.
3. It may give you warnings during the first few iterations. This is due to the inability to drive the residual to 0 when the parameters are still way off. Such warnings are normal and should not be a concern.

4. `optimize.leastsq` returns an array. The first element in the array is the optimal value of the parameter you are optimizing over. Therefore, you want to invoke it like this:
`opt_n = optimize.leastsq(opt_n, n_init, args=(...))[0][0]`
5. Solve the problem several times with several initial values of n to see if it gives a better solution.

Problem 2 deliverables: Your code should print the optimum value of the ideality parameter n , and plot a single graph with the $\log(\text{Diode Current})$ on the y-axis and the source voltage on the x-axis. The graph should have two curves; the measured and the calculated diode current. If you estimated the ideality parameter accurately, you should get coinciding curves.

Make sure to submit a single file named `project3.py` containing both codes.

MANDATORY – Reminder to cite your AI chat. Check the syllabus document for more details